

# EXAMINATION QUESTION PAPER

## Paper: Set A

Total Questions: 4

Total Marks: 7

Date: 09-01-2026

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### Q1. [1 marks] [Easy] [data structures]

Question: Which fundamental data structure organizes elements in a sequential, linear manner, where each element is directly linked to the next?

- A) Tree
- B) Graph
- C) Hash Table
- D) Array

Correct Answer: D

### Q2. [2 marks] [Medium] [data structures]

Question: Which fundamental data structure is most appropriate for implementing a Breadth-First Search (BFS) algorithm on a graph, owing to its First-In, First-Out (FIFO) access pattern?

- A) Stack
- B) Linked List
- C) Priority Queue
- D) Queue

Correct Answer: D

### Q3. [3 marks] [Hard] [data structures]

Question: Consider a scenario where an in-memory database index requires efficient retrieval of records based on a composite key  $(K_1, K_2)$ . The operations performed most frequently are exact-match lookups on  $K_1$  and range queries on  $K_2$  for a specific  $K_1$ . Which data structure offers the optimal worst-case time complexity for \*both\* these primary operations under these specific conditions?

- A) A B+-Tree where  $K_1$  is the primary key and  $K_2$  is indexed hierarchically within each  $K_1$  node.
- B) A standard Hash Table where the key is the concatenation of  $K_1$  and  $K_2$ .
- C) A 2D Range Tree optimized for orthogonal range queries.
- D) A Radix Tree (Trie) indexed first by  $K_1$  and then by  $K_2$  using pointer redirection upon key collision.

Correct Answer: A

### Q4. [1 marks] [Easy] [dbms]

Question: Which of the following is the primary function of a Database Management System (DBMS)?

- A) To create graphical user interfaces for applications
- B) To manage and control access to stored data
- C) To compile source code into executable machine code
- D) To manage network connectivity between servers

Correct Answer: B